

Ex.No: 11

Formulae:

i. To compute y_{in} for an activation function:

$$y_{in} = b + \sum_i^n x_i w_{ij}$$

ii. Sigmoid activation function:

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

iii. Mean Squared Error (MSE):

$$\sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

sum of the errors of all samples

Note: In our simulation, we used the same bias for every node; however, in practice, biases are often adjusted individually to allow for more flexibility in learning complex patterns.

The ANN architecture consists of input, hidden, and output layers, while the hidden layers perform weighted computations and apply activation functions.

Observations (Simulation Screenshots):

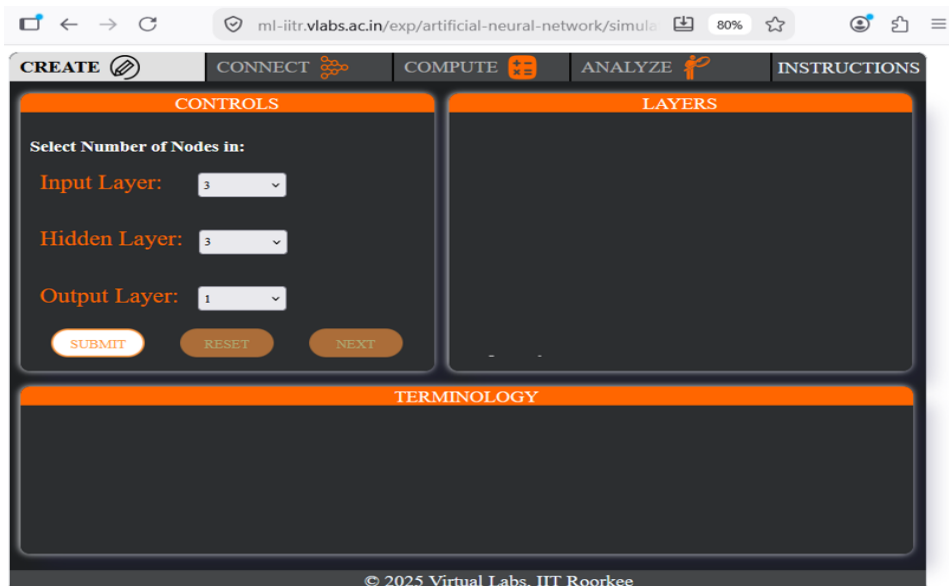


Fig 1: Selecting the number of nodes for the input, hidden, and output layers

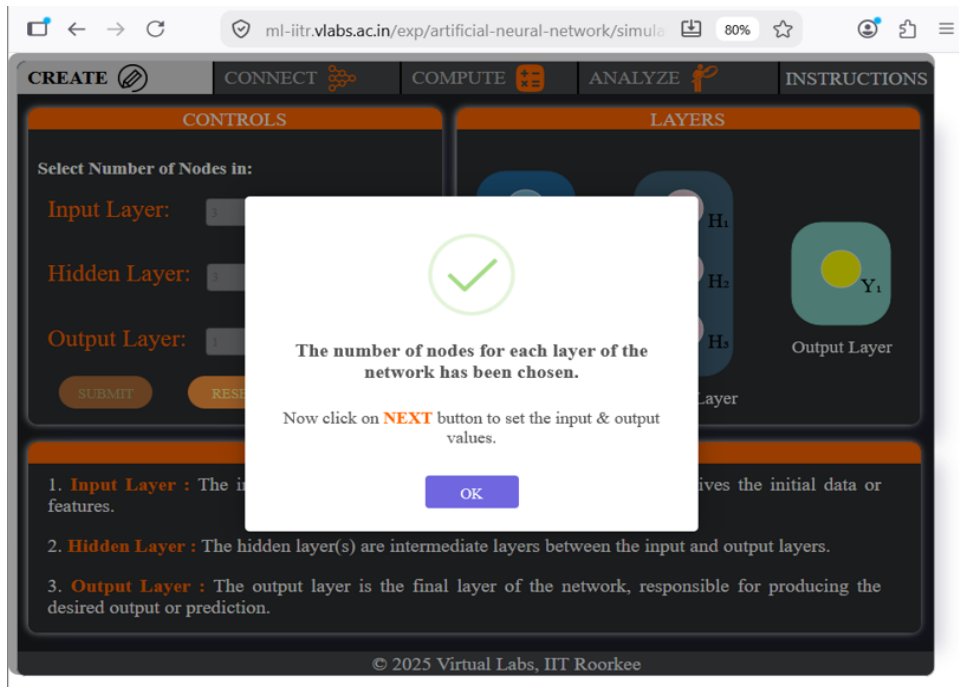


Fig 2: Confirmation message after the number of nodes is chosen

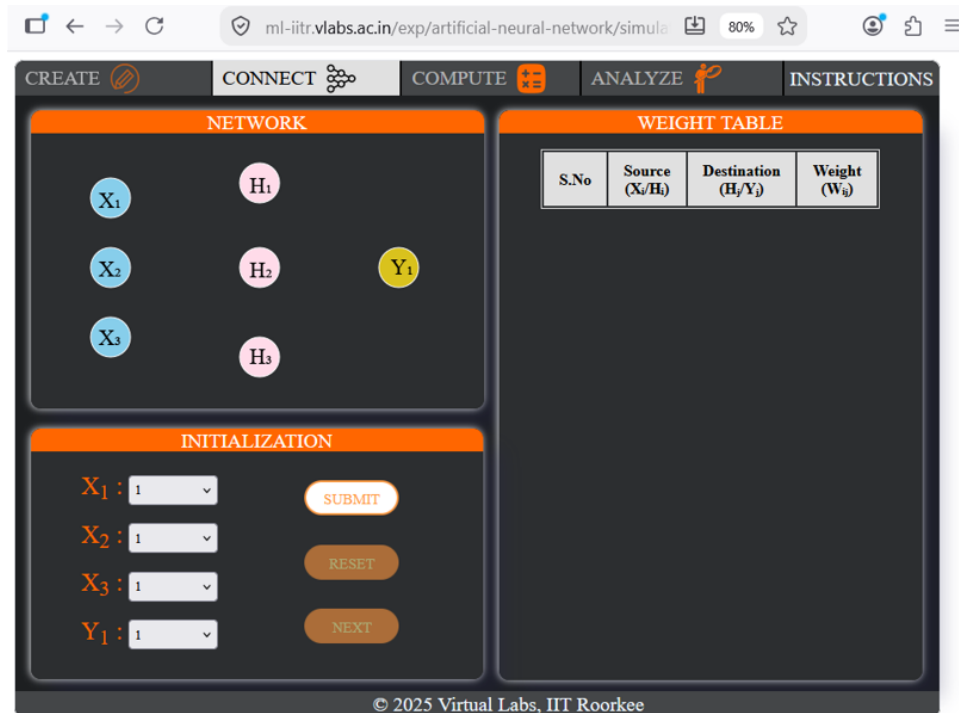


Fig 3: Network layout with fields to initialize input (X_i) and output (Y_i) values

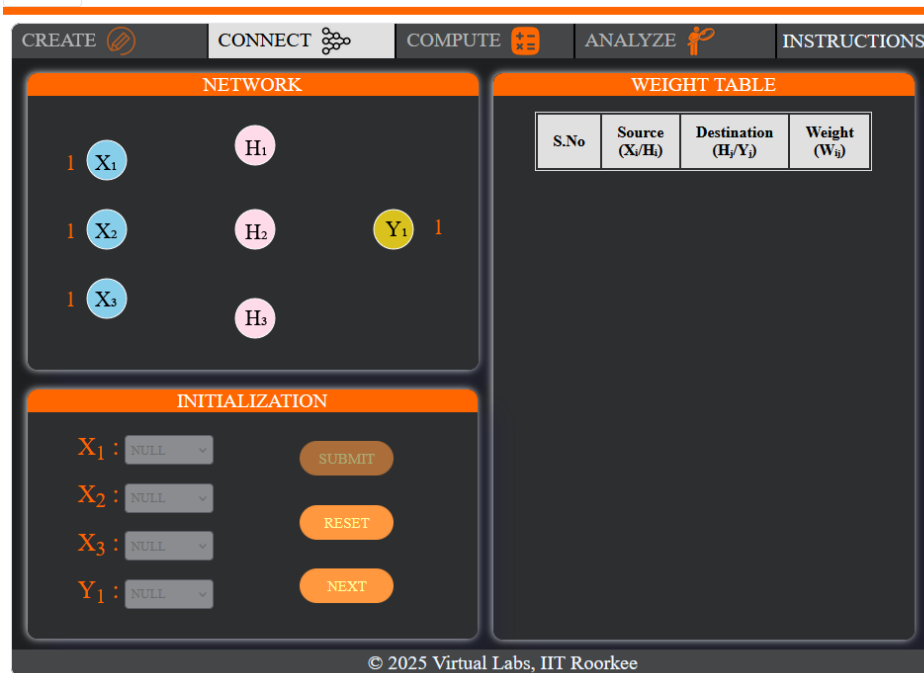


Fig 4: Input and output values submitted ($X_1 = X_2 = X_3 = 1, Y_1 = 1$)

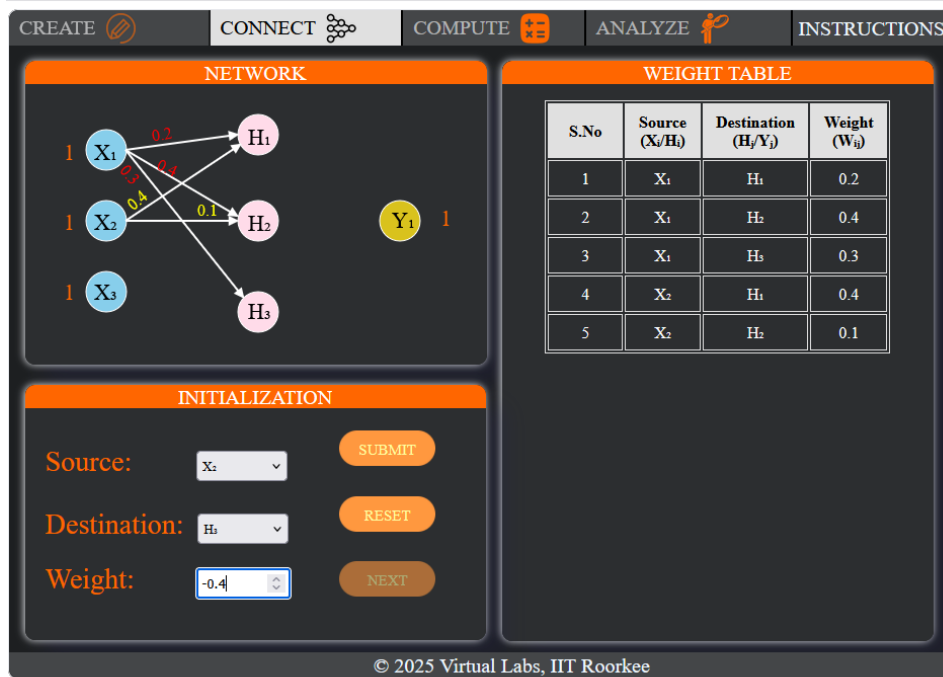


Fig 5: Assigning a weight between an input node and a hidden node ($X_2 \rightarrow H_2 = -0.4$)

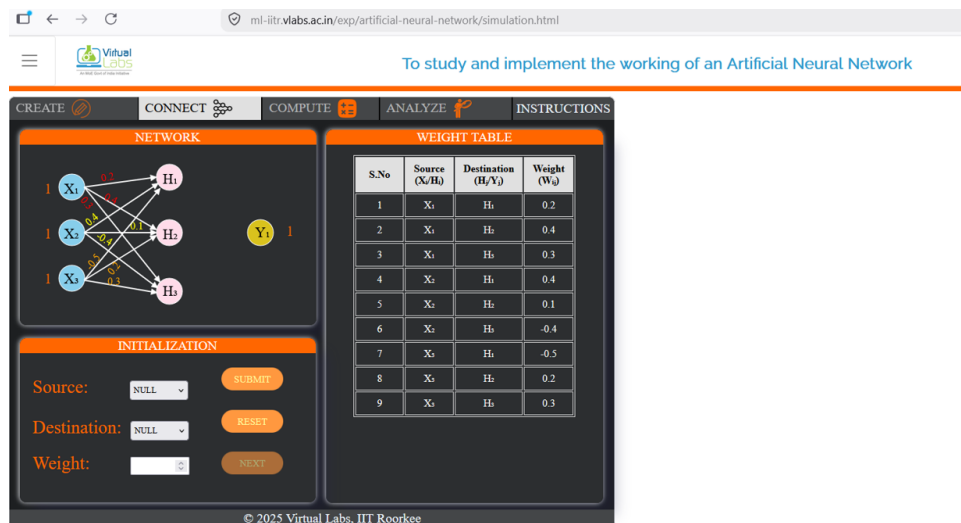


Fig 6: Weight table after all input-hidden layer connections are assigned

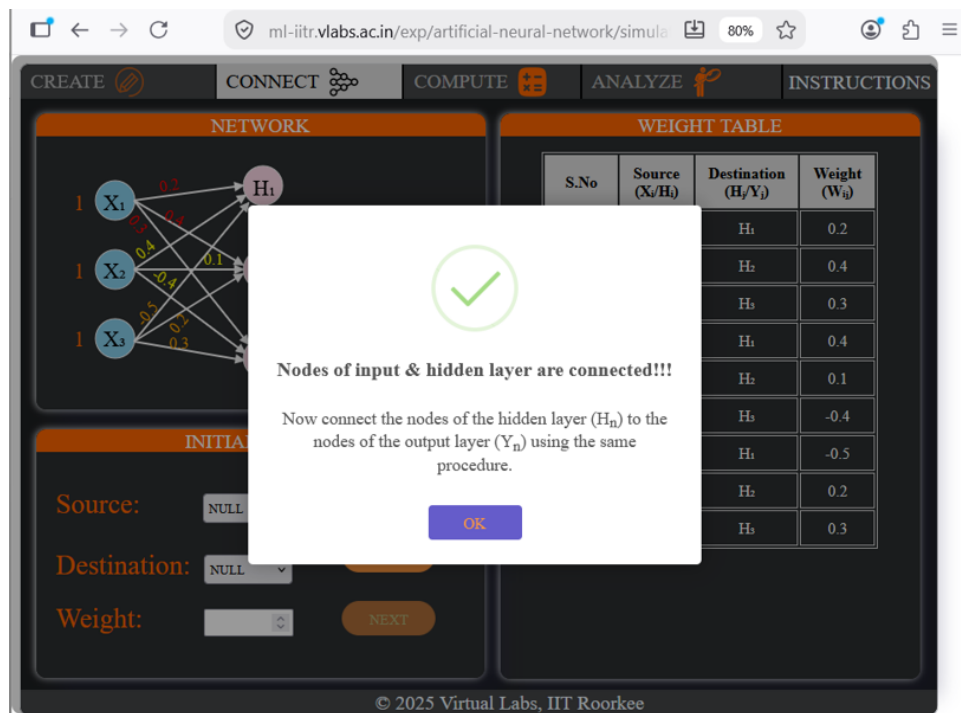


Fig 7: Confirmation – input and hidden layer nodes connected

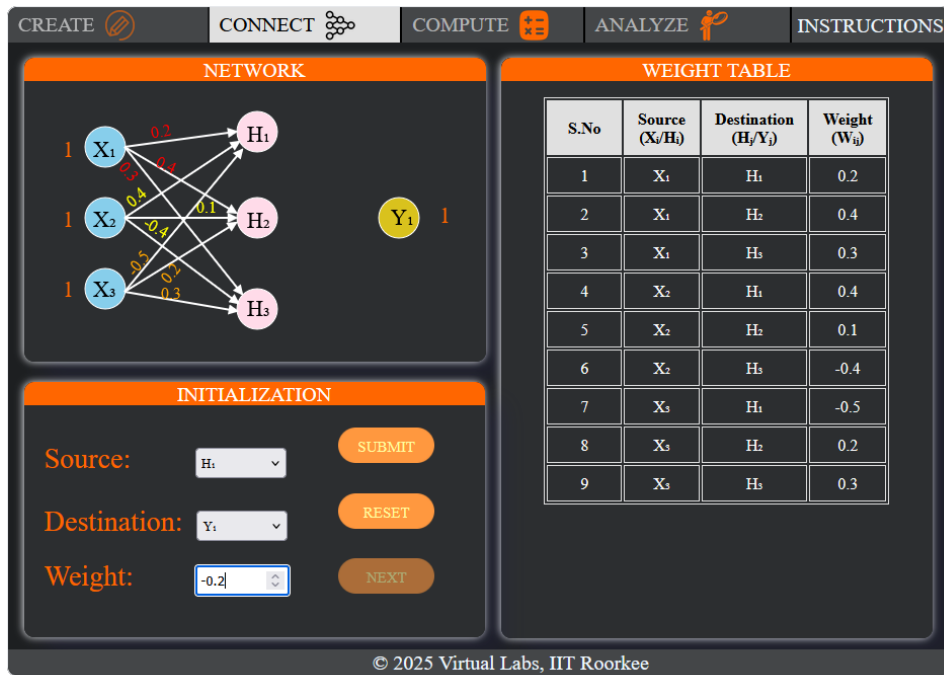


Fig 8: Assigning a weight between a hidden node and the output node ($H_1 \rightarrow Y_1 = -0.2$)

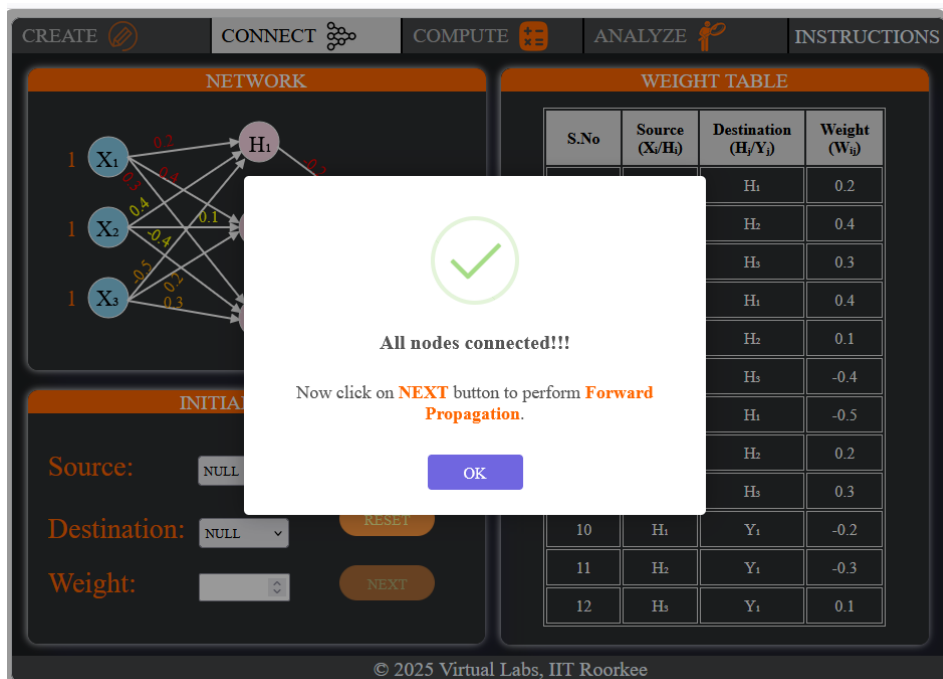


Fig 9: Confirmation – all nodes connected, complete weight table (12 connections)

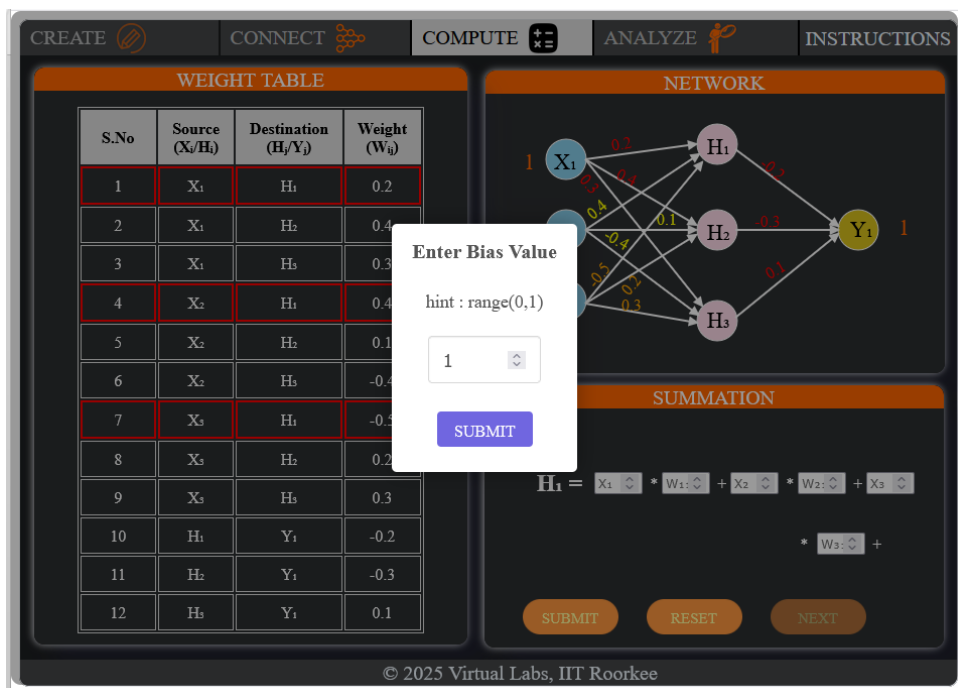


Fig 10: Entering the bias value

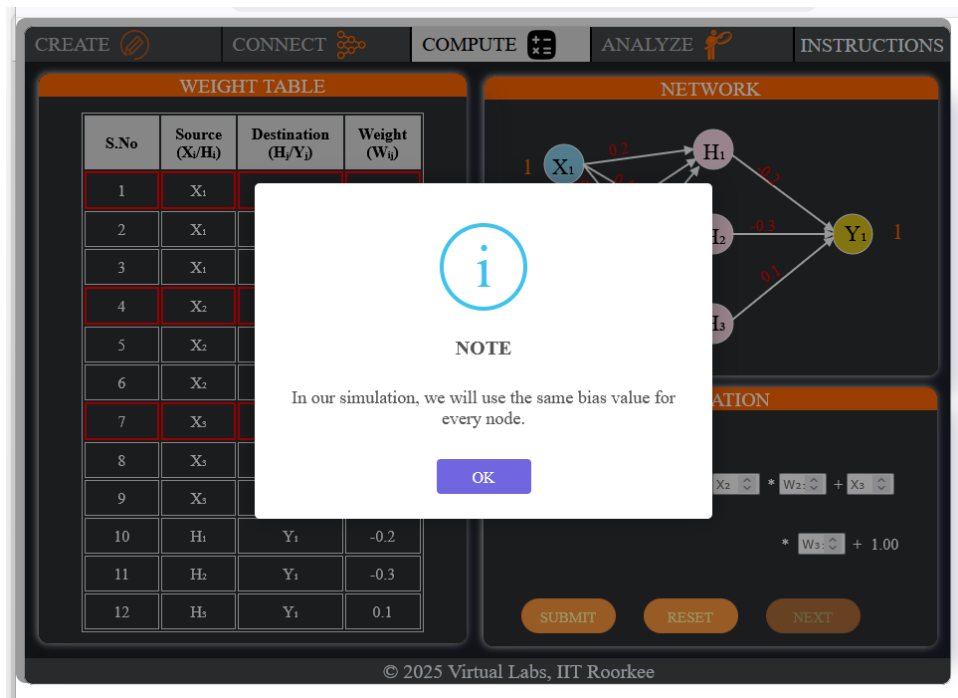


Fig 11: The simulation uses the same bias value for every node

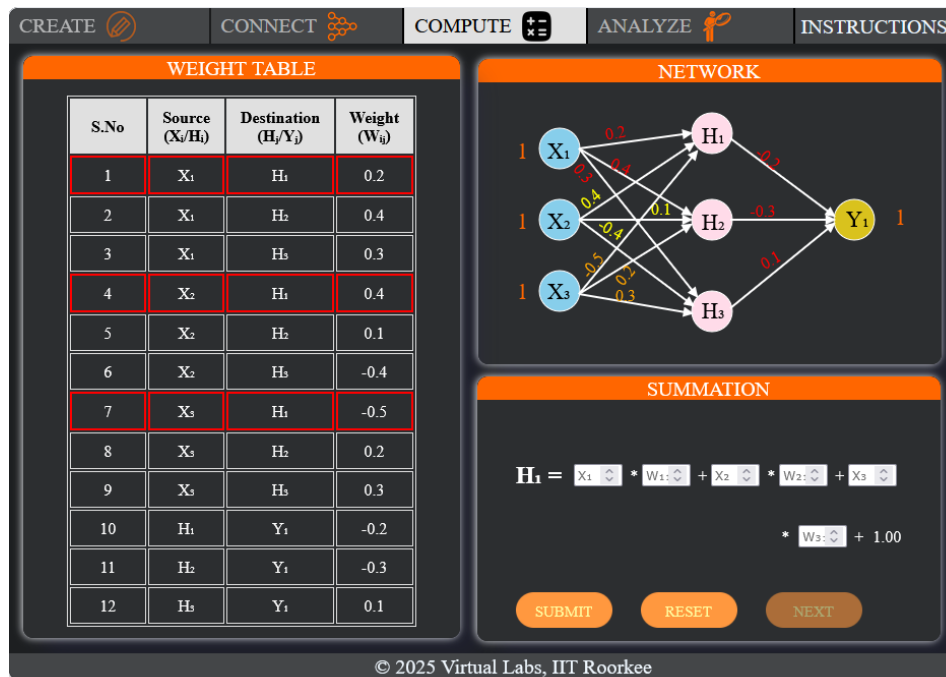


Fig 12: Summation setup for hidden layer node H1

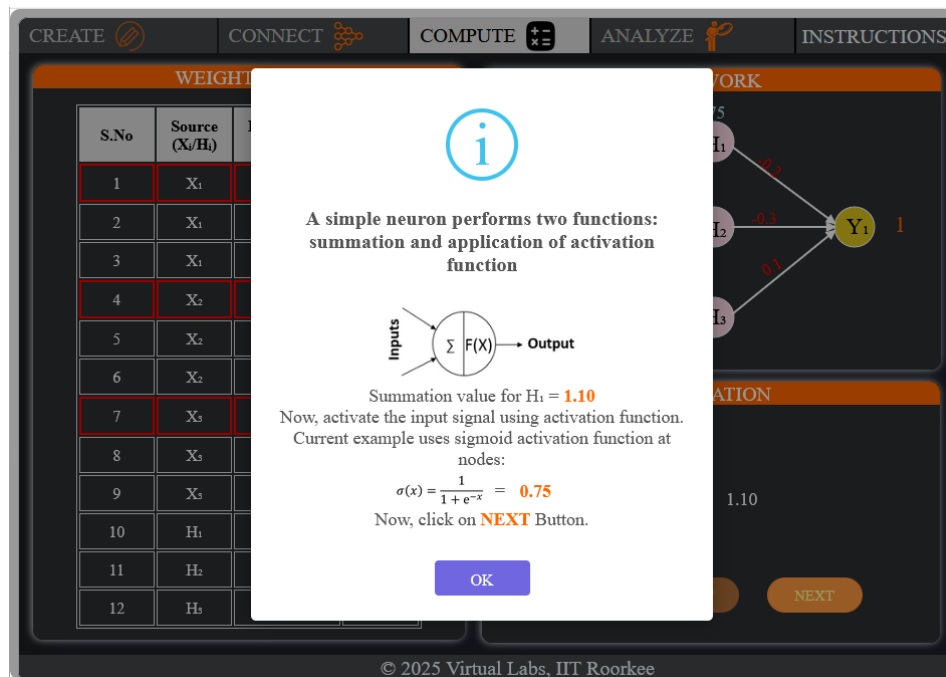


Fig 13: Summation (1.10) and sigmoid activation (0.75) computed for H1

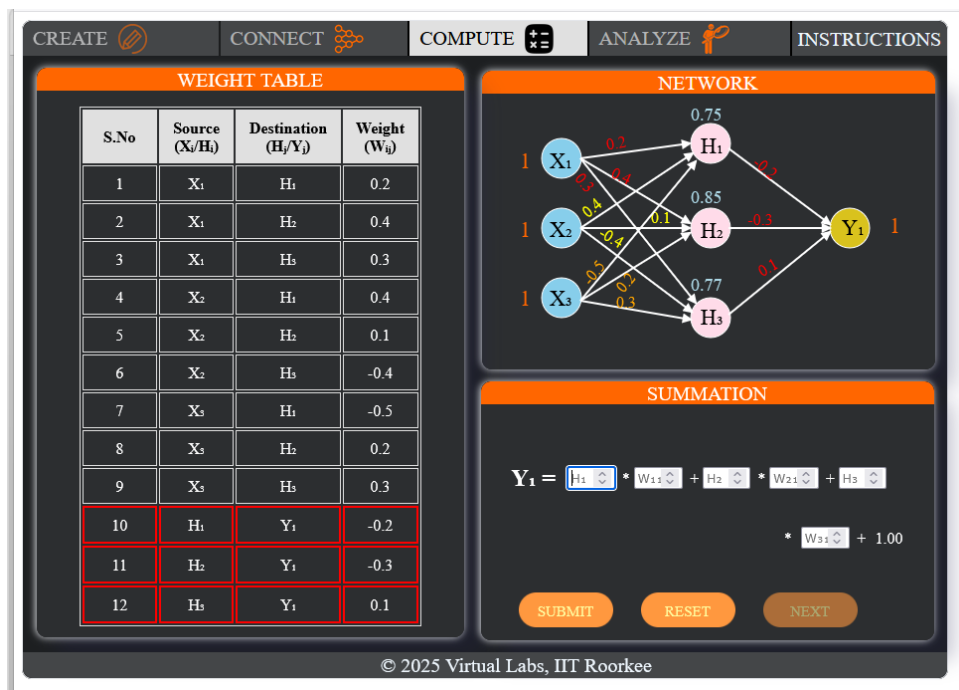


Fig 14: Hidden layer outputs obtained ; summation setup for output node Y1

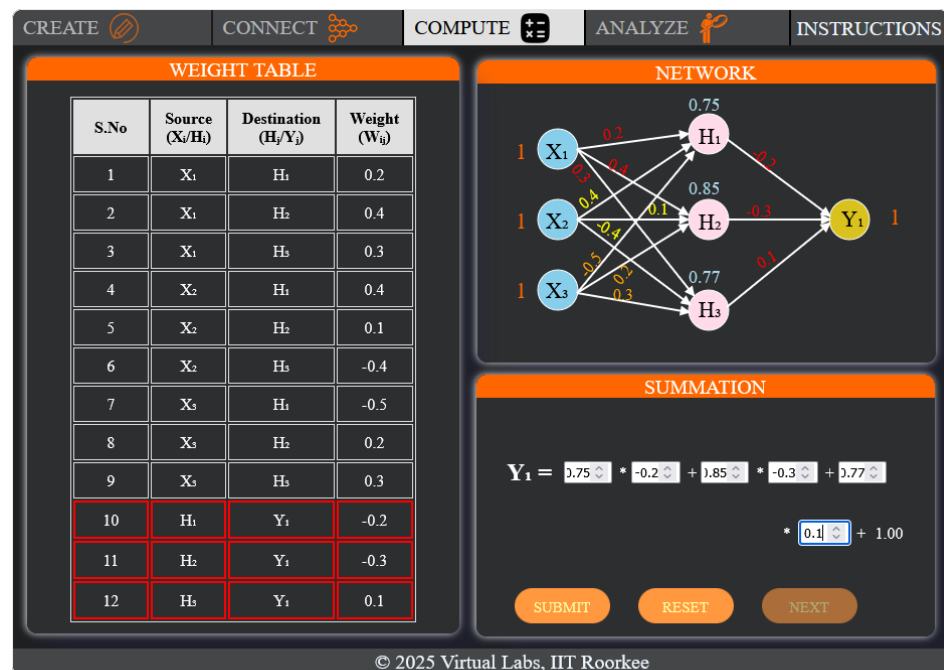


Fig 15: Output layer summation values entered for Y1

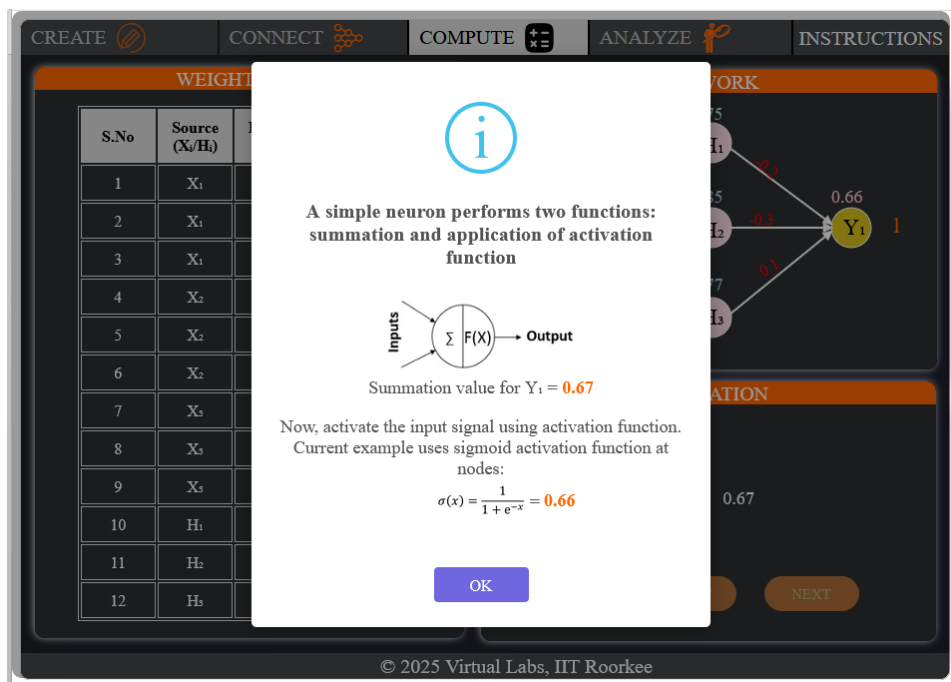


Fig 16: Summation (0.67) and sigmoid activation (0.66) computed for Y₁

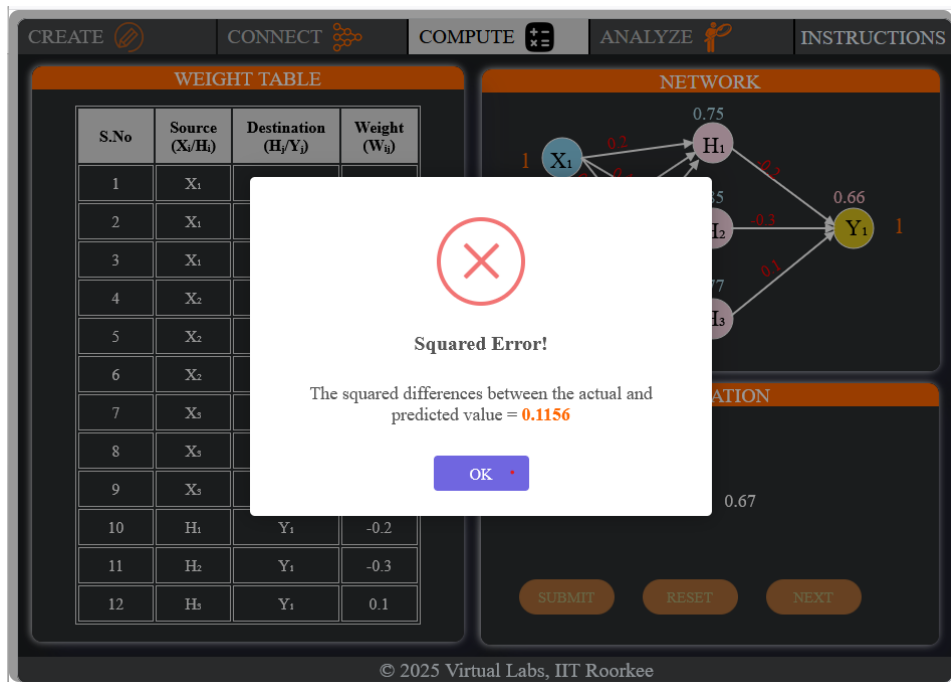


Fig 17: Squared error computed between actual and predicted output (0.1156)

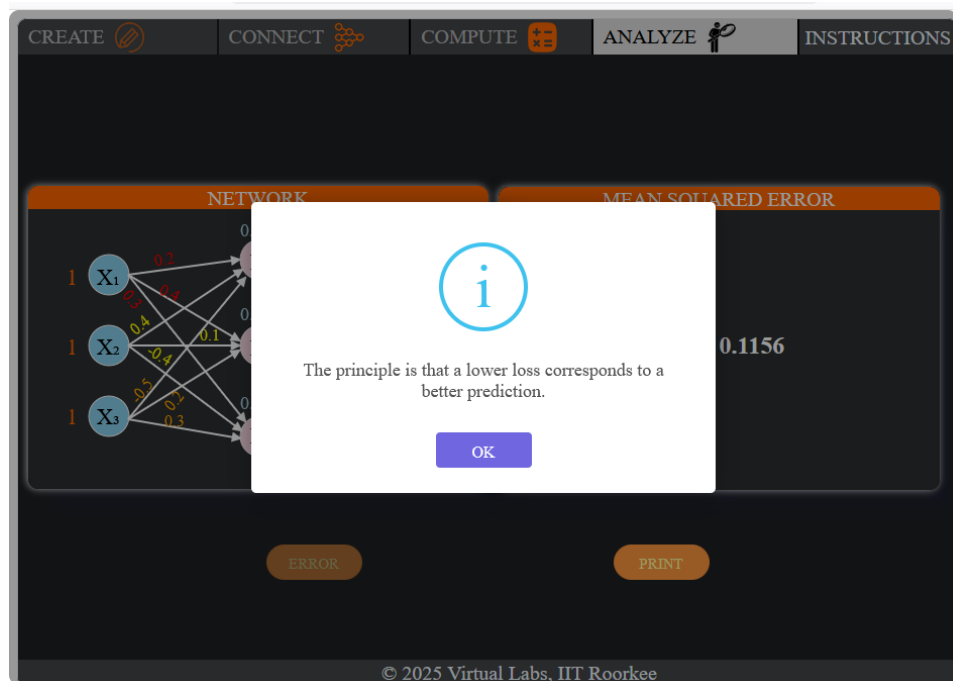


Fig 18: A Lower Loss corresponds to a better prediction

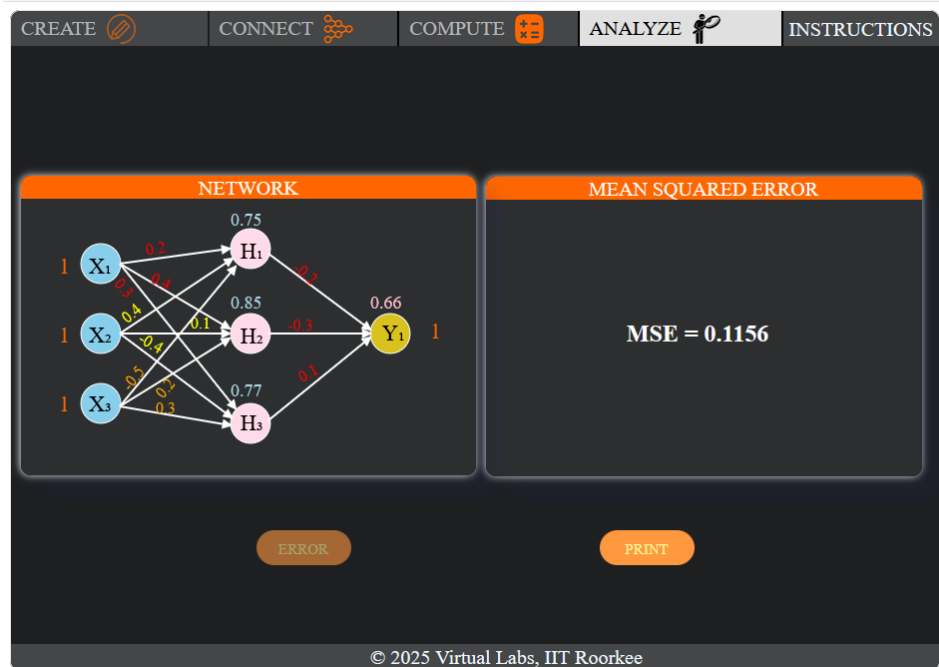


Fig 19: Final network with computed node outputs and Mean Squared Error (MSE = 0.1156)

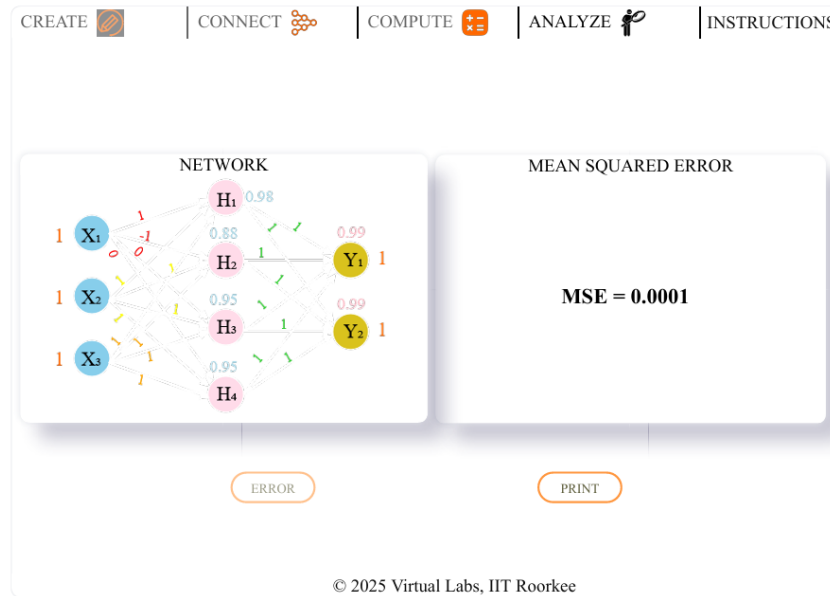


Fig 20: Printable summary view of the network and Mean Squared Error

Simulation link: <https://ml-iitr.vlabs.ac.in/exp/artificial-neural-network/simulation.html>